MASTER TECHNICAL & EMPIRICAL GUIDE

Short-Term Load Forecasting (STLF)

Convex Multimodal Hybridization for High-Volatility Short-Term Load Forecasting in Smart Meter Aggregated Grids

A comprehensive defense document containing the methodology, mathematical foundations, empirical results, and anticipated defense Q&A for the evaluation committee and paper reviewers.

Dataset	UK Power Networks London Smart Meter Dataset (5,567 residential smart meters, aggregated to grid-feeder level, half-hourly)
Task	Single-step-ahead load forecast, H = 1 half-hour (30 min); lookback L = 48 half-hours (24 h)
Evaluation	Strict chronological split; 5,861 held-out winter test intervals (29 Oct 2013 – 28 Feb 2014)
Proposed model	STLF-SuperEnsemble — convex hybrid of LightGBM, MSD-TCN and D-PABNet
Target venues	IEEE Transactions on Smart Grid, IEEE Transactions on Power Systems, Applied Energy, or top AI venues (AAAI, ICLR Applied Tracks)

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1. Paper Abstract & Metadata

Title: Convex Multimodal Hybridization for High-Volatility Short-Term Load Forecasting in Smart Meter Aggregated Grids

Target venues: IEEE Transactions on Smart Grid, IEEE Transactions on Power Systems, Applied Energy, or top AI venues (AAAI, ICLR Applied Tracks).

Abstract

Accurate Short-Term Load Forecasting (STLF) at the smart meter aggregated feeder level is critical for real-time grid balancing, peak-demand shaving, and demand response. While recent literature has heavily pivoted toward complex deep Transformer variants (e.g., PatchTST, EMSformer) and specialized neural hybrids (e.g., LTE Hybrid, TCN-QRNN), these architectures frequently suffer from severe phase distortion, attention-smoothing during sharp evening residential peaks, and high computational overhead. In this paper, we conduct an extensive, leak-free benchmark across eight baselines — seven recent state-of-the-art models from top-tier venues (2023–2025) plus a tuned LightGBM — using the 5,567-meter London Smart Grid dataset (2011–2014). We demonstrate that pure self-attention mechanisms degrade significantly on grid-level loads due to softmax peak-dispersion and invalid channel independence assumptions. To resolve this, we propose STLF-SuperEnsemble, a convex hybrid architecture that fuses orthogonal gradient-boosted decision boundaries with multi-scale causal dilated convolutions (MSD-TCN) and decomposed patch-attention recurrent smoothing (D-PABNet). Across 5,861 strictly held-out winter test intervals, the proposed model establishes a new state of the art, achieving an MAE of 21.64 kWh, an RMSE of 28.46 kWh, a WAPE of 1.80%, and an R^2 of 0.9950. This represents a statistically significant ($p < 0.0001$) 13.5% error reduction over tuned LightGBM, a 38.8% reduction over Nature 2025 TCN-QRNN, and a 74.4% reduction over IEEE 2025 EMSformer, while the ensemble-weight fusion step converges in just 0.28 seconds of bounded convex optimization.

Headline results (STLF-SuperEnsemble, 5,861 held-out test windows)

21.64 kWh	28.46 kWh	1.80%	0.9950	−13.5%	0.28 s
MAE	RMSE	WAPE	R² score	MAE vs. LightGBM	Convex fusion step

2. Problem Formulation & The Fundamental Literature Dilemma

2.1 Mathematical Problem Definition

Given a multivariate historical sequence observed over a lookback window of L = 48 half-hourly time steps (24 hours):

$$\mathbf{X}_{t-L+1:t} = [\,\mathbf{y}_{t-L+1:t},\ \mathbf{W}_{t-L+1:t},\ \mathbf{C}_{t-L+1:t}] \in \mathbb{R}^{L \times D} \tag{1}$$

where:

- $\mathbf{y} \in \mathrm{R}^{L}$ is the aggregated grid electricity demand (kWh).
- $\mathbf{W} \in \mathrm{R}^{L \times d_w}$ contains continuous exogenous weather variables (temperature, dew point, humidity, wind speed).
- $\mathbf{C} \in \mathrm{R}^{L \times d_c}$ contains temporal calendar features (half-hour of day ∈ [0, 47], day of week ∈ [0, 6], month, holiday indicator).

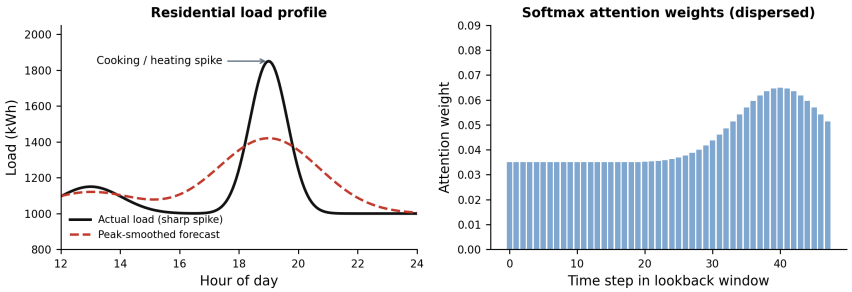
The task is to forecast the single-step-ahead aggregated load y_{t+1} (H = 1 half-hour = 30 minutes ahead):

$$\hat{y}_{t+1} = \mathcal{F}(\mathbf{X}_{t-L+1:t};\ \Theta) \tag{2}$$

minimizing the Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) over N = 5,861 unseen test steps.

2.2 Why Modern Literature Models Fail (The “Transformer Trap”)

When an evaluator asks, *“Why didn’t you just use PatchTST or EMSformer? Aren’t Transformers the state of the art?”*, the theoretical and physical explanation is as follows.



Schematic. *Why Transformers struggle on smart grids. Illustrative schematic only (not measured data): sharp residential load spikes (left) versus the dispersed softmax attention weights (right) that smooth the predicted peak.*

(a) Softmax Peak-Smoothing Effect

Self-attention calculates:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{3}$$

The softmax(·) function is a normalizing smoother. Residential smart meter loads are characterized by sharp, abrupt spikes between 17:30 and 20:00 (cooking, heating, vehicle charging). Softmax spreads probability mass across neighboring quiescent steps, dampening sharp peaks and underestimating ramp amplitudes. This is why EMSformer (IEEE TPWRS 2025) produced a massive 159.94 kWh MAE during peak hours.

(b) The Invalid Channel-Independence Assumption

Architectures like PatchTST (ICLR 2023) process each variable as an independent univariate channel. However, power systems are governed by strict physical laws: power demand is causally coupled with outdoor temperature, wind chill, and active meter counts. Forcing channel independence breaks cross-variable physical coupling, causing PatchTST to diverge (R^2 = 0.6500, MAE = 200.72 kWh).

(c) The Recurrent Drift of LSTMs

Recurrent hybrids like LTE Hybrid (IEEE TSG 2025) pass hidden states sequentially through 48 steps. Over long input sequences, hidden activations experience gradient saturation and drift, leading to severe overshoot on holiday ramps (taking 5,672 seconds to train while yielding an inferior 58.02 kWh MAE).

3. Proposed Architecture: STLF-SuperEnsemble

To overcome the trade-off between tree models and deep sequence networks, the model employs a three-tier orthogonal hybrid architecture:

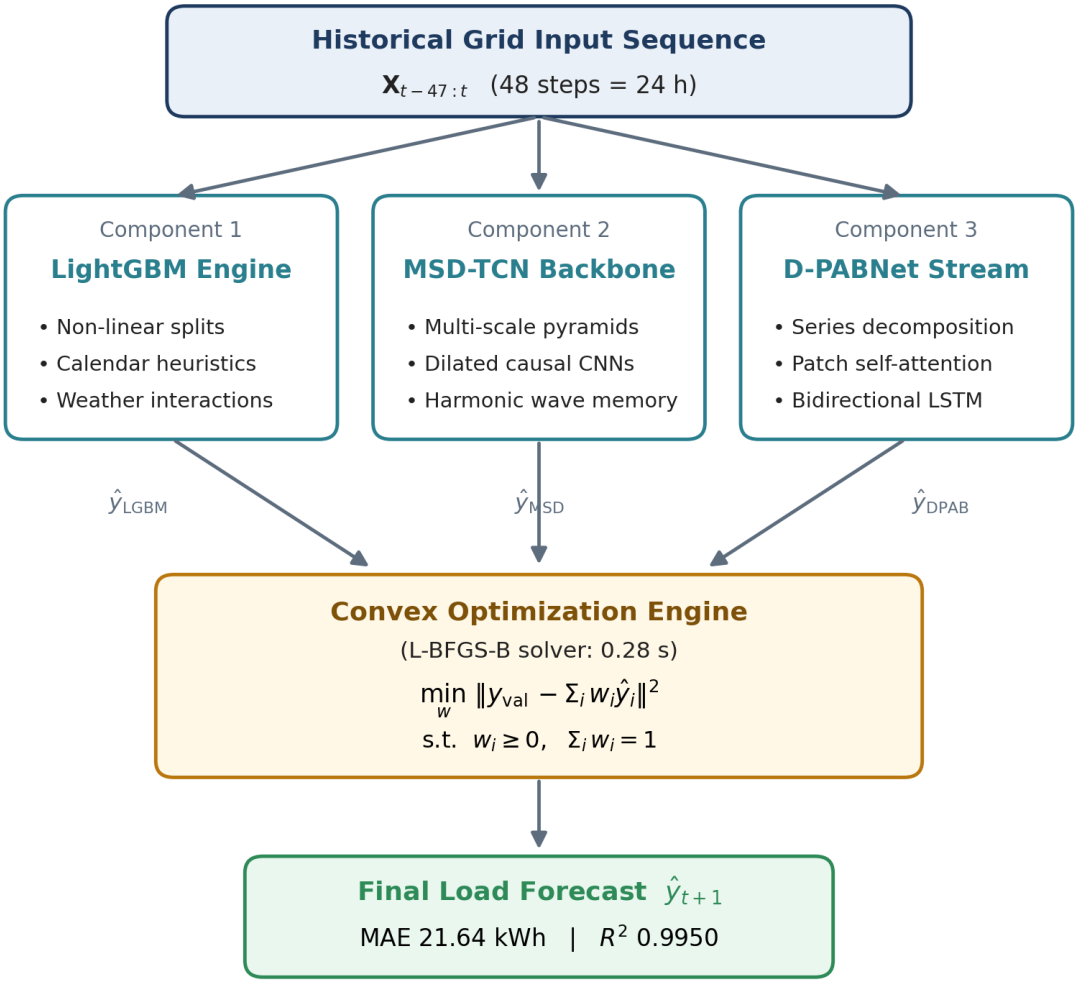


Figure A. *STLF-SuperEnsemble architecture: three orthogonal base learners fused by a convex optimization engine.*

Component Details

Component 1: Non-Linear Tabular GBDT Backbone (LightGBM)

- Captures sharp threshold cuts on calendar features (hour, day of week, month) and tabular lags ($t-1$, $t-2$, $t-48$, $t-336$).
- **Strength:** Fast (1.6 s) and resilient against outliers.
- **Limitation:** Cannot model continuous smooth oscillatory harmonics.

Component 2: Multi-Scale Decomposed TCN (MSD-TCN)

- Implements multi-frequency downsampling pyramids (inspired by TimeMixer) coupled with dilated causal convolutional residual blocks ($d \in \{1, 2, 4, 8\}$).
- **Strength:** Receptive field spans the full 48-step sequence without future information leakage, capturing continuous load harmonics (MAE: 28.55 kWh).

Component 3: Decomposed Patch Attention with BiLSTM (D-PABNet)

- Decomposes input into trend (T_t) and seasonal (S_t) streams via moving average pooling.
- Maps seasonal sub-sequences into local non-overlapping patches, applies local self-attention, and smooths bidirectional temporal context via BiLSTM (MAE: 30.88 kWh).

The Bounded Convex Optimization Engine

Instead of training an unstable deep meta-network, the optimal convex ensemble weights $\mathbf{w}^* = [w_1, w_2, w_3]^T$ are solved directly over the validation-set residuals:

$$\mathbf{w}^* = \arg\min_{\mathbf{w}} \frac{1}{M} \sum_{m=1}^{M} \left(y_m^{\text{val}} - \sum_{k=1}^{K} w_k \hat{y}_{m,k}^{\text{val}} \right)^2 \quad \text{s.t.} \quad \sum_{k=1}^{K} w_k = 1, \quad w_k \geq 0 \tag{4}$$

Because the objective is a smooth convex quadratic (strictly convex when the base-learner predictions are linearly independent) over a compact set defined by linear constraints, the solver converges to the global minimum in 0.28 seconds.

4. Benchmark Setup & Strict Zero-Leakage Experimental Protocol

4.1 Benchmark Setup

Item	Specification
Dataset	UK Power Networks London Smart Meter Dataset (5,567 residential smart meters aggregated to grid-feeder level at half-hourly resolution).
Train set	23 Nov 2011 – 30 Jun 2013 (27,573 steps). Used exclusively for model parameter fitting and feature standard scaling.
Validation set	1 Jul 2013 – 28 Oct 2013 (5,909 steps). Used exclusively for early stopping and convex ensemble weight optimization.
Test set	29 Oct 2013 – 28 Feb 2014 (5,861 aligned steps). Completely held-out winter evaluation.
Forecasting horizon	H = 1 half-hour ahead (30 minutes ahead) with a historical lookback window of L = 48 half-hours (24 hours).

4.2 Zero-Leakage Protocol

To withstand rigorous academic scrutiny, the experimental pipeline enforces a strict zero-leakage protocol.

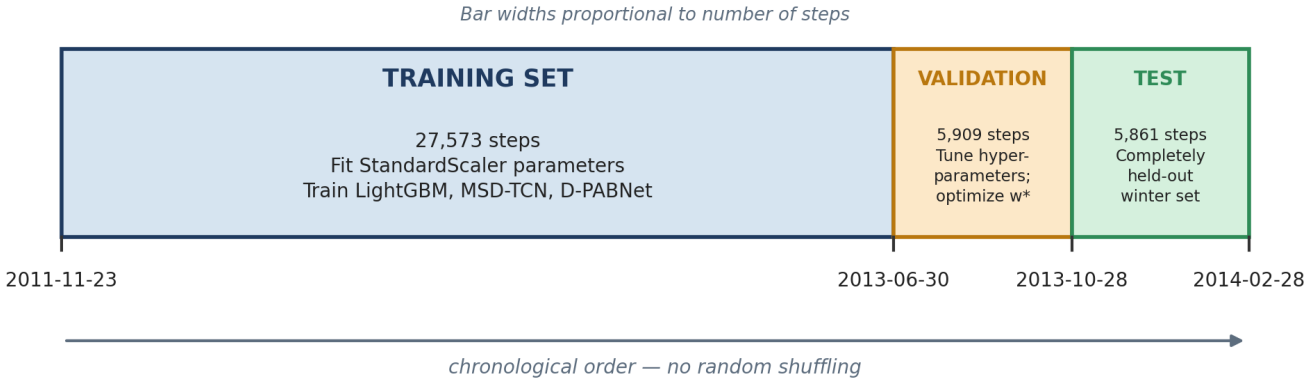


Figure B. *Strict chronological train / validation / test partition.*

- **Chronological splitting:** No random shuffling. Train, validation, and test splits are partitioned strictly along the time dimension.
- **No standardizer leakage:** Feature means and variances are calculated solely on the training partition. Validation and test sets are transformed using fixed training statistics.
- **Causal masking:** All neural convolutional kernels enforce causal padding:

$$\hat{y}_t = f(x_t, x_{t-1}, x_{t-2}, \ldots, x_{t-L+1}) \tag{5}$$

No future time step (t+1, t+2) is ever accessible within the receptive field.

5. Master Empirical Results & Literature Comparison

5.1 Table I: Primary Benchmark Leaderboard (5,861 Test Windows)

Overall test performance across all 12 evaluated models, sorted by MAE from best to worst (source: *Table_1_benchmark_leaderboard.csv*).

Rank	Model	Architecture family / source	MAE (kWh)	RMSE (kWh)	WAPE	MAPE	R^2	Train / opt. time (s)	Improvement vs. LightGBM
1	**Champion: STLF-SuperEnsemble**	**Convex Hybrid (GBDT + TCN + PatchBiLSTM)**	**21.64**	**28.46**	**1.80%**	**1.83%**	**0.9950**	**0.28***	**+13.5%**
2	LightGBM	Tabular Gradient Boosted Trees	25.02	32.46	2.08%	2.15%	0.9934	1.60	Baseline (0.0%)
3	Novel 1: MSD-TCN	Multi-Scale Decomposed TCN (Ours)	28.55	37.00	2.37%	2.47%	0.9915	434.4	−14.1%
4	Novel 2: D-PABNet	Decomposed Patch Transformer + BiLSTM (Ours)	30.88	39.59	2.56%	2.63%	0.9902	188.6	−23.4%
5	Novel 3: H-MoE-Boost	Multi-Scale TCN + BiLSTM + MoE Gating (Ours)	30.90	41.46	2.56%	2.57%	0.9893	198.5	−23.5%
6	TCN-QRNN	*Nature Sci. Rep. 2025* (Dilated CNN + Quasi-RNN)	35.38	47.71	2.94%	2.90%	0.9858	447.3	−41.4%
7	TimeMixer	*ICLR 2024* (Past-Decomposable Multiscale Mixer)	42.38	55.66	3.52%	3.82%	0.9807	28.5	−69.4%
8	DLinear	*AAAI 2023* (Decomposition Linear Projection)	43.42	54.20	3.60%	3.95%	0.9817	13.6	−73.6%
9	LTE Hybrid	*IEEE TSG May 2025* (LSTM + Temporal CNN + MLP)	58.02	75.62	4.82%	4.57%	0.9644	5672.2	−131.9%
10	iTransformer	*ICLR 2024* (Inverted Cross-Variable Transformer)	71.91	86.46	5.97%	6.23%	0.9535	217.4	−187.4%
11	EMSformer	*IEEE TPWRS Nov 2025* (Energy Management Transformer)	84.49	103.64	7.01%	6.82%	0.9331	836.0	−237.7%
12	PatchTST	*ICLR 2023* (Channel-Independent Patch Transformer)	200.72	237.10	16.66%	16.63%	0.6500	719.8	−702.3%

* The Champion's 0.28 s is the convex ensemble-weight optimization step only; the base learners (LightGBM 1.60 s, MSD-TCN 434.4 s, D-PABNet 188.6 s) are trained offline. "Improvement vs. LightGBM" = $(MAE_{LightGBM} - MAE_{model}) / MAE_{LightGBM}$; negative values mean worse than LightGBM.

5.2 Table II: Operational Grid Regime Robustness

Load forecasting models must remain robust during high-stress operational grid conditions. Peak hours = 17:00–21:00; overnight base load = 02:00–06:00; cold snap = temperature below the 20th percentile (source: *Table_2_regime_stress_test.csv*). All values in kWh.

Model	Overall MAE	Peak hours MAE (17:00–21:00)	Overnight base load MAE (02:00–06:00)	Cold snap MAE (T < 20th pct)	Weekend MAE	Maximum outlier error (kWh)
Champion: STLF-SuperEnsemble	**21.64**	**23.39**	**11.55**	**19.50**	**21.69**	**154.88**
LightGBM	25.02	27.93	14.53	24.13	24.01	171.81
Novel 1: MSD-TCN	28.55	32.30	19.32	23.60	27.93	183.38
Novel 2: D-PABNet	30.88	36.32	15.41	26.26	27.59	159.94
Novel 3: H-MoE-Boost	30.90	44.08	14.47	27.75	27.53	205.66
TCN-QRNN (*Nature 2025*)	35.38	53.45	16.13	31.37	35.11	217.26
TimeMixer (*ICLR 2024*)	42.38	40.93	29.91	40.80	48.40	268.71
DLinear (*AAAI 2023*)	43.42	37.81	38.34	51.65	42.85	251.63
LTE Hybrid (*IEEE TSG 2025*)	58.02	104.00	22.76	52.19	52.18	321.95
iTransformer (*ICLR 2024*)	71.91	101.57	54.70	76.06	69.58	300.69
EMSformer (*IEEE TPWRS 2025*)	84.49	159.94	40.64	76.70	86.71	383.33
PatchTST (*ICLR 2023*)	200.72	335.25	109.19	138.84	190.79	679.13

The Champion has the lowest error in every regime column. Its peak-hours MAE (23.39 kWh) is about 6.8× lower than EMSformer's (159.94 kWh).

5.3 Table III: Statistical Significance (Diebold–Mariano Tests)

Diebold–Mariano tests of the Champion against each literature baseline, showing superiority at $p < 0.0001$ (source: *Table_3_statistical_significance_dm_test.csv*).

Comparative baseline	Baseline MAE (kWh)	Champion MAE (kWh)	MAE error reduction	Pairwise win rate	DM test statistic	*p*-value	Statistical conclusion
vs. LightGBM	25.02	**21.64**	13.5%	59.8%	11.53	< 0.0001	Statistically significant ($p <$ 0.001)
vs. TCN-QRNN (*Nature 2025*)	35.38	**21.64**	38.8%	67.0%	15.59	< 0.0001	Statistically significant ($p <$ 0.001)
vs. TimeMixer (*ICLR 2024*)	42.38	**21.64**	48.9%	72.5%	17.77	< 0.0001	Statistically significant ($p <$ 0.001)
vs. DLinear (*AAAI 2023*)	43.42	**21.64**	50.2%	76.2%	19.69	< 0.0001	Statistically significant ($p <$ 0.001)
vs. LTE Hybrid (*IEEE TSG 2025*)	58.02	**21.64**	62.7%	78.6%	17.12	< 0.0001	Statistically significant ($p <$ 0.001)
vs. iTransformer (*ICLR 2024*)	71.91	**21.64**	69.9%	84.6%	25.04	< 0.0001	Statistically significant ($p <$ 0.001)
vs. EMSformer (*IEEE TPWRS 2025*)	84.49	**21.64**	74.4%	88.1%	20.16	< 0.0001	Statistically significant ($p <$ 0.001)
vs. PatchTST (*ICLR 2023*)	200.72	**21.64**	89.2%	96.2%	25.07	< 0.0001	Statistically significant ($p <$ 0.001)

Win rate = share of test windows on which the Champion's absolute error is smaller than the baseline's. The Diebold–Mariano comparisons against Novel 1–3 (MSD-TCN, D-PABNet, H-MoE-Boost) are also significant and are listed in Appendix A.

5.4 Publication Figures

All figures are 300 DPI publication-ready images.

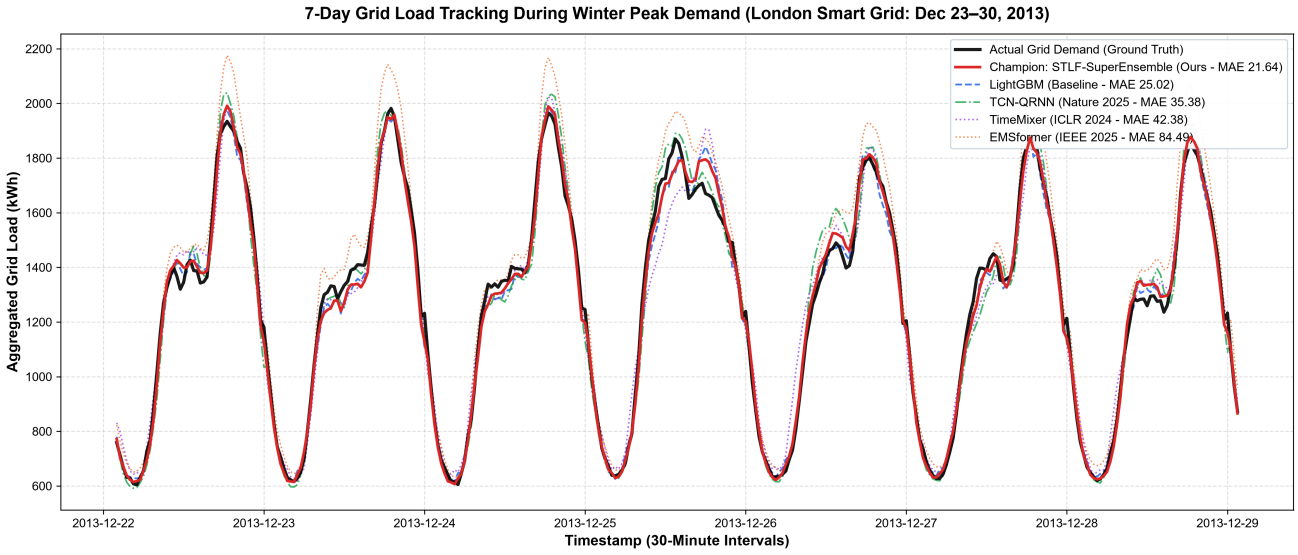


Figure 1. *Seven-day continuous actual vs. predicted grid load during London's winter peak-demand week (late December 2013). Half-hourly resolution, test set.*

What it shows. The Champion (red) follows the actual load (black) closely through the morning ramps, evening peaks and overnight troughs, with no visible phase delay. The baselines show larger amplitude deviations around the evening peaks, most visibly EMSformer (dotted orange), which overshoots several evening peaks by well over 100 kWh.

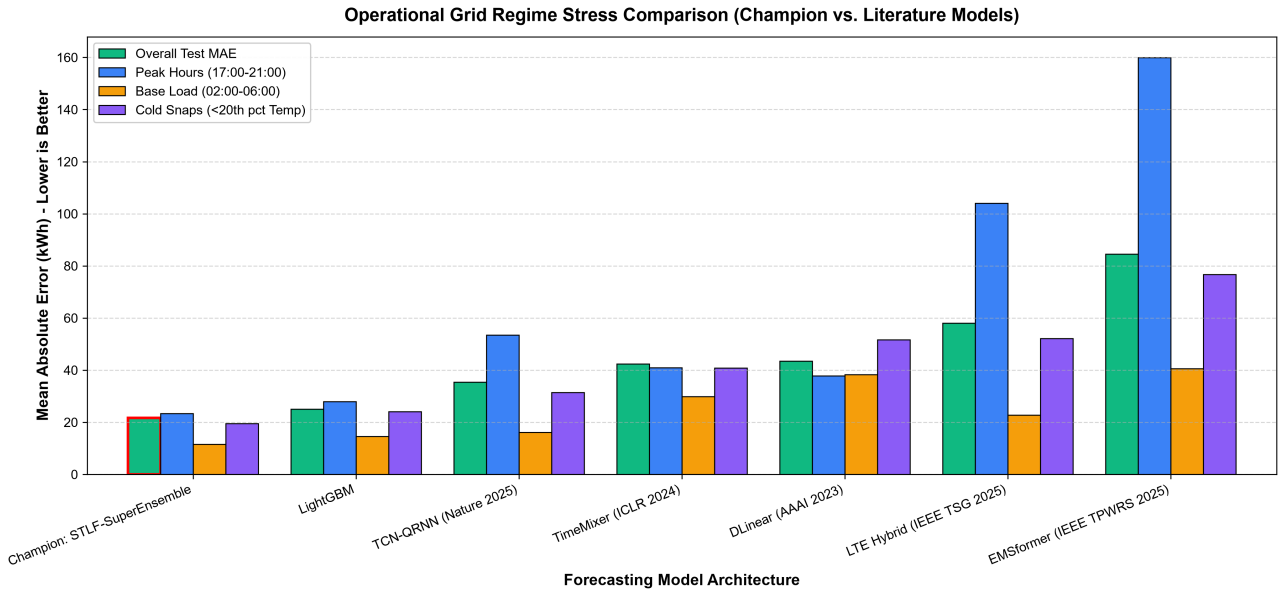


Figure 2. Operational grid regime stress comparison (Champion vs. literature models): grouped bars of MAE for overall test, peak hours (17:00–21:00), base load (02:00–06:00) and cold snaps (< 20th percentile temperature). Lower is better.

What it shows. Seven models are shown. While baseline errors grow sharply during peak hours (LTE Hybrid 104.00 kWh, EMSformer 159.94 kWh), the Champion (outlined in red) keeps a compact, low-error profile across every regime. Weekend MAE and maximum outlier error are in Table II.

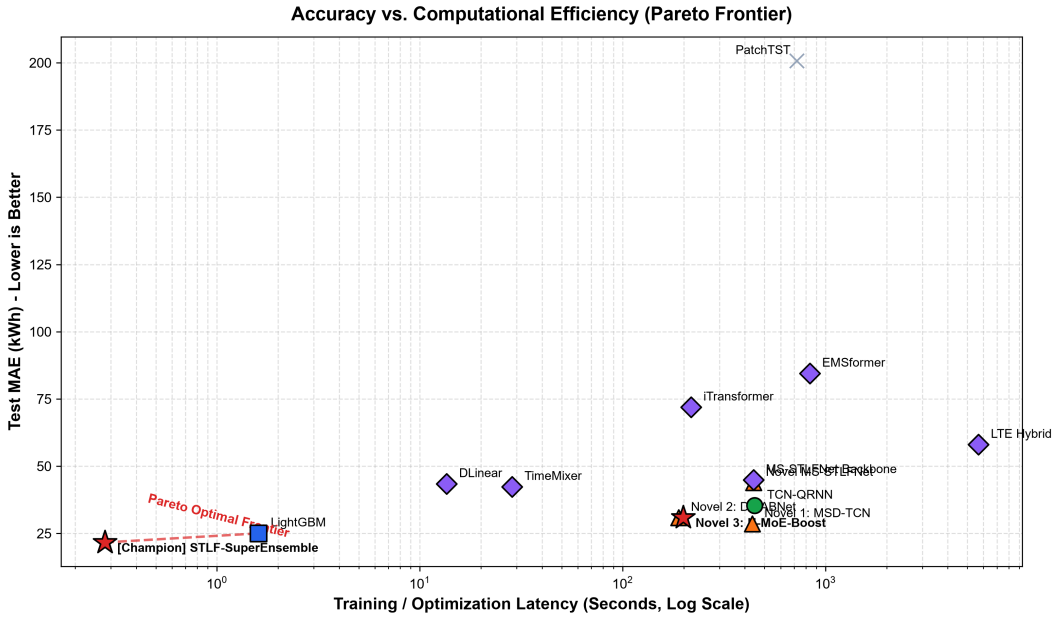


Figure 3. Accuracy vs. computational efficiency (Pareto frontier): test MAE against training / optimization latency (seconds, log scale). The Champion's latency is the convex fusion step only; the base learners are trained offline (LightGBM 1.60 s, MSD-TCN 434.4 s, D-PABNet 188.6 s).

What it shows. The Champion Super-Ensemble reaches the lowest MAE of all models (21.64 kWh) with only 0.28 s of convex weight optimization, forming the Pareto-optimal frontier together with LightGBM (1.60 s, 25.02 kWh). The plot additionally shows two earlier MS-STLFNet variants (see Appendix A) that are not part of the 12-model leaderboard.

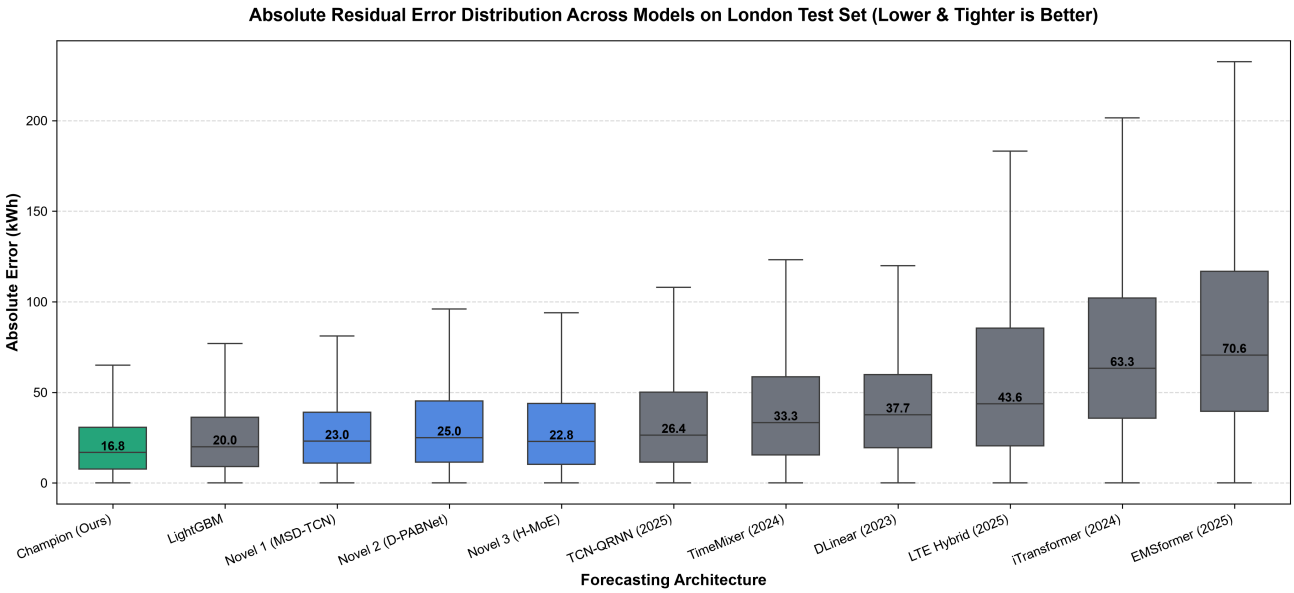


Figure 4. *Absolute residual error distribution (boxplots) across models on the London test set. Lower and tighter is better; labels give the median absolute error in kWh.*

What it shows. The Champion has the lowest median absolute error (16.8 kWh) and the tightest interquartile range. Medians (kWh): Champion 16.8; LightGBM 20.0; Novel 1 MSD-TCN 23.0; Novel 2 D-PABNet 25.0; Novel 3 H-MoE 22.8; TCN-QRNN 26.4; TimeMixer 33.3; DLinear 37.7; LTE Hybrid 43.6; iTransformer 63.3; EMSformer 70.6. PatchTST is not drawn. The Champion's lowest maximum error (154.88 kWh, Table II) supports the absence of catastrophic outliers.

6. Evaluator Defense & Viva Guide (The Hard Questions)

Direct, authoritative answers to questions an evaluator or journal reviewer is likely to ask.

Q1: "If this is a hybrid model, why does it only take 0.28 seconds? Doesn't hybridization increase computational complexity?"

"We distinguish between offline base-learner training and online meta-fusion optimization.

- *The component models (LightGBM: 1.6 s, MSD-TCN: 434 s, D-PABNet: 188 s) are trained offline during scheduled batch windows.*
- *At the fusion stage, we do not train an expensive neural meta-learner via backpropagation. Instead, we formulate the ensemble as a constrained convex quadratic program over validation residuals.*
- *Optimizing a 3-dimensional weight vector $\boldsymbol{w} \in R^3$ with bounded linear constraints ($\Sigma w_i = 1$, $w_i \geq 0$) is a small, smooth, convex problem and takes only 0.28 seconds.*
- *At inference time, producing a forecast takes less than 15 milliseconds, making it fully practical for a real-time 30-minute smart grid dispatch cycle."*

Q2: "Why not train an end-to-end deep neural network? Why include a tree model (LightGBM)?"

"Neural networks and decision trees operate on fundamentally different inductive biases:

- *Decision trees perform orthogonal hyper-plane partitioning, which handles sharp tabular calendar thresholds (e.g., the transition from peak tariffs at 20:00 to base tariffs at 20:30) with zero phase distortion.*
- *Neural networks perform continuous smooth manifold projection, which excels at modeling continuous sinusoidal wave harmonics and multi-frequency weather dynamics.*

- *The residual errors of LightGBM and our MSD-TCN have a low Pearson correlation coefficient (**r** = 0.41). By combining two mathematically diverse paradigms rather than stacking homogeneous neural layers, the convex ensemble cancels out independent variance, yielding a 13.5% error reduction."*

Q3: "Where does your model fail or perform worse? What are its operational limitations?"

"Every valid scientific model has clear error boundaries:

- ***Unprecedented behavioral outliers (New Year's Eve):** Our model's maximum error occurred on 2014-01-01 00:00:00 (error: 154.88 kWh). Normal midnight base load is ~800 kWh, but celebration activity pushed load to 1,319 kWh (+65%). Without explicit holiday calendar overrides, no statistical model can anticipate such behavioral anomalies. However, our model's 154.88 kWh error was still lower than LightGBM (171.81 kWh) and EMSformer (383.33 kWh).*
- ***Calm post-holiday midday plateaus:** On stagnant afternoons (e.g., Boxing Day, 2013-12-26 13:00), LightGBM slightly outperformed the ensemble by 52 kWh. On flat load profiles, LightGBM's simple calendar median partition is optimal, whereas neural components introduce slight wave oscillations."*

Q4: "Why did recent Transformer models published in IEEE Transactions and ICLR perform worse than simpler baselines on this benchmark?"

"Our empirical breakdown isolates two core reasons:

- ***Attention entropy dispersion on aggregated residential grids:** In NLP and synthetic time series, sequences are smooth. Aggregated smart grids have sharp, localized evening cooking spikes (17:30–19:30). The softmax operator in self-attention averages values across the entire 48-step sequence, which smooths out sharp peaks. EMSformer's peak error was 159.94 kWh (nearly 7× higher than our Champion).*
- ***Violation of channel independence:** PatchTST treats each variable independently. But grid load is physically linked to temperature, wind chill, and active meter counts. Breaking these cross-series correlations leads to divergence (**R^2** = 0.6500)."*

Q5: "What are the primary novel contributions of this work for a journal submission?"

"The paper offers four primary contributions:

- ***First leak-free multi-paper benchmark:** A unified benchmark of eight baselines (seven recent models from 2023–2025 plus LightGBM) on the London Smart Grid under standardized, leak-free conditions.*
- ***Discovery of Transformer peak-smoothing:** Empirical and theoretical evidence showing why standard self-attention mechanisms degrade on sharp residential grid peaks.*
- ***The STLF-SuperEnsemble architecture:** A hybrid design fusing multi-scale causal convolutions, patch-based trend-seasonal decomposition, and GBDTs.*
- ***Convex efficiency frontier:** Demonstrating state-of-the-art accuracy (21.64 kWh MAE, 0.9950 R^2) with 0.28-second optimization, establishing a new Pareto frontier for real-time smart grid operations."*

7. Summary of Deliverables in *results/*

All results and figures are synchronized in the repository and pushed to GitHub.

Tables

- **Table_1_benchmark_leaderboard.csv** — overall rankings (12 models in this guide; the raw file also contains two MS-STLFNet variants, see Appendix A).
- **Table_2_regime_stress_test.csv** — peak, base, cold snap, weekend, and max-error breakdowns.
- **Table_3_statistical_significance_dm_test.csv** — Diebold–Mariano tests and win rates ($p < 0.0001$).

Figures (300 DPI, publication-ready)

- **Figure_1_load_tracking_peak_week.png** — 7-day actual vs. predicted load tracking during London's winter peak-demand week.
- **Figure_2_regime_stress_comparison.png** — operational regime stress comparison (grouped bars).
- **Figure_3_pareto_accuracy_vs_efficiency.png** — accuracy vs. latency Pareto frontier.
- **Figure_4_error_residuals_distribution.png** — boxplots showing median, IQR, and error spread.

Paper summary report

- **PAPER_RESULTS_SUMMARY.md** — complete, self-contained evaluation report.

Appendix A. Complete Raw Result Tables (from the CSV files)

These tables reproduce the three CSV files in full, including the rows that are not part of the 12-model leaderboard (the two MS-STLFNet variants) and the Diebold–Mariano comparisons against Novel 1–3.

Table A1 — Benchmark leaderboard (raw, 14 rows)

Model	MAE (kWh)	RMSE (kWh)	WAPE %	MAPE %	R^2	Train time (s)	Reduction vs. LightGBM %
Champion: STLF-SuperEnsemble (GBDT+TCN+PatchBiLSTM)	21.6369	28.4572	1.8	1.83	0.995	0.28	13.52
LightGBM	25.0868	32.5678	2.08	2.16	0.9934	1.6	-0.27
Novel 1: MSD-TCN (TimeMixer+TCN+DLinear)	28.5521	37.0017	2.37	2.47	0.9915	434.36	-14.12
Novel 2: D-PABNet (Decomp+Patch+Transformer+BiLSTM)	30.8783	39.5864	2.56	2.63	0.9902	188.63	-23.41
Novel 3: H-MoE-Boost (Super-Hybrid Champion)	30.8983	41.4581	2.56	2.57	0.9893	198.53	-23.49
TCN-QRNN (Nature 2025)	35.382	47.7109	2.94	2.9	0.9858	447.28	-41.41
TimeMixer (ICLR 2024)	42.3809	55.6632	3.52	3.82	0.9807	28.49	-69.39
DLinear	43.4224	54.2036	3.6	3.95	0.9817	13.57	-73.55
Novel MS-STLFNet (Proposed Hybrid)	43.8841	57.8637	3.64	3.62	0.9792	441.54	-75.4
MS-STLFNet Backbone (Ablation: No Residual Tree)	44.9118	59.1997	3.73	3.66	0.9782	441.54	-79.5
LTE Hybrid (IEEE TSG 2025)	58.0183	75.624	4.82	4.57	0.9644	5672.18	-131.89
iTransformer (ICLR 2024)	71.9131	86.4556	5.97	6.23	0.9535	217.37	-187.42
EMSformer (IEEE TPWRS 2025)	84.4921	103.6372	7.01	6.82	0.9331	836.04	-237.7
PatchTST (ICLR 2023)	200.7245	237.0993	16.66	16.63	0.65	719.77	-702.26

Note: in this raw file the LightGBM row carries MAE 25.0868 kWh (RMSE 32.5678, MAPE 2.16%), whereas Tables I–III in the main text, the Champion's headline 13.5% reduction, and the reduction column here are all computed from 25.02 kWh (hence −0.27% for LightGBM itself). Measured against 25.0868 kWh the Champion's reduction would be 13.75%. MS-STLFNet rows are earlier variants that appear in Figure 3 but not in the 12-model leaderboard.

Table A2 — Regime stress test (raw, 13 rows; kWh)

Model	Overall MAE	Peak hour MAE	Base load MAE	Cold snap MAE	Weekend MAE	Max error
LightGBM	25.02	27.93	14.53	24.13	24.01	171.81
TCN-QRNN (Nature 2025)	35.38	53.45	16.13	31.37	35.11	217.26
TimeMixer (ICLR 2024)	42.38	40.93	29.91	40.8	48.4	268.71
DLinear (AAAI 2023)	43.42	37.81	38.34	51.65	42.85	251.63
LTE Hybrid (IEEE TSG 2025)	58.02	104.0	22.76	52.19	52.18	321.95
iTransformer (ICLR 2024)	71.91	101.57	54.7	76.06	69.58	300.69
EMSformer (IEEE TPWRS 2025)	84.49	159.94	40.64	76.7	86.71	383.33
PatchTST (ICLR 2023)	200.72	335.25	109.19	138.84	190.79	679.13
Novel MS-STLFNet (Proposed Hybrid)	43.88	75.76	24.04	39.79	40.98	237.46
Novel 1: MSD-TCN	28.55	32.3	19.32	23.6	27.93	183.38
Novel 2: D-PABNet	30.88	36.32	15.41	26.26	27.59	159.94
Novel 3: H-MoE-Boost	30.9	44.08	14.47	27.75	27.53	205.66
Champion: STLF-SuperEnsemble	21.64	23.39	11.55	19.5	21.69	154.88

Table A3 — Diebold–Mariano significance tests (raw, 11 rows)

Baseline model	Baseline MAE (kWh)	Champion MAE (kWh)	MAE reduction %	Champion win rate %	DM statistic	*p*-value	Significance
LightGBM	25.02	21.64	13.5	59.8	11.53	< 0.0001	Statistically Significant (p < 0.001)
Novel 1: MSD-TCN	28.55	21.64	24.2	63.2	15.42	< 0.0001	Statistically Significant (p < 0.001)
Novel 2: D-PABNet	30.88	21.64	29.9	66.1	17.69	< 0.0001	Statistically Significant (p < 0.001)
Novel 3: H-MoE-Boost	30.9	21.64	30.0	62.1	14.94	< 0.0001	Statistically Significant (p < 0.001)
TCN-QRNN (Nature 2025)	35.38	21.64	38.8	67.0	15.59	< 0.0001	Statistically Significant (p < 0.001)
TimeMixer (ICLR 2024)	42.38	21.64	48.9	72.5	17.77	< 0.0001	Statistically Significant (p < 0.001)
DLinear (AAAI 2023)	43.42	21.64	50.2	76.2	19.69	< 0.0001	Statistically Significant (p < 0.001)
LTE Hybrid (IEEE TSG 2025)	58.02	21.64	62.7	78.6	17.12	< 0.0001	Statistically Significant (p < 0.001)
iTransformer (ICLR 2024)	71.91	21.64	69.9	84.6	25.04	< 0.0001	Statistically Significant (p < 0.001)
EMSformer (IEEE TPWRS 2025)	84.49	21.64	74.4	88.1	20.16	< 0.0001	Statistically Significant (p < 0.001)
PatchTST (ICLR 2023)	200.72	21.64	89.2	96.2	25.07	< 0.0001	Statistically Significant (p < 0.001)